

Tarcísio G. Rodrigues

Senior Software Engineer

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Overview

Backend software engineer with strong expertise in low-latency, high-performance critical systems, having worked with market data applications, payment applications and risk analysis software.

Employment

- 2026 **Software Engineer**, *Cetacea*, São Paulo, Remote
Designed and developed .NET (C#) backend services for handling customer tickets lifecycle and middleware interfacing with Matrix communication service. Built and maintained a React/TypeScript/GraphQL frontend layer, adding a dedicated realtime widget for speeding up initial customer onboarding. Advised on adopting spec-driven development (SDD) practices, authoring the initial spec templates and establishing the review workflow.
- 2025–2026 **Software Engineer**, *Mottu*, São Paulo, Hybrid
Built .NET (C#) backend services to integrate with both Santander PIX Payment infrastructure and with BBVA México payment services.
- 2024–2025 **Software Engineer**, *mParticle*, New York, Remote
Contributed to the evolution and support of a distributed querying system for extracting marketing performance analytics (audience segmentation, funnel analysis) from large-scale customer event data. Primarily developed backend services in Java, with additional contributions to the JavaScript frontend.
- 2016–2023 **Software Engineer**, *XP Inc.*, São Paulo
Implemented software for market data feed in C++ with Boost (ASIO, Signals2, Multi-Index, ICL, etc.), QuickFIX, QuickFAST and SBE reference implementations. Experience with market data protocols, FIX/FAST and SBE encodings, and B3's UMDF platform. Designed and developed software infrastructure to integrate backend and frontend services, employing .NET micro-services, Kafka, MongoDB, Docker, Kubernetes, WebSockets, etc.
- 2014–2016 **Software Engineer**, *FIT – Institute of Technology*, Sorocaba
Developed C++ library for payment applications. CMake and Python scripting for building automation. Experience with the EMV standard and Verifone's SDK.
- 2014 **Software Engineer**, *MAPS – Financial Market Solutions*, São Paulo
Developed and maintained risk analysis software in C++ with Microsoft Foundation Classes. Continuous integration system maintenance and development with Jenkins.
- 2013–2014 **Software Engineer**, *IGS – International Gaming Solutions*, São Paulo
Developed embedded backend for land-based bingo games with boost. Port of legacy C code to Linux. Developed continuous integration system on Buildbot. Experience with IGT's SAS (Slot Accounting System) protocol.
- 2012–2013 **Software Analyst**, *FiberWork – Optical Communications*, Campinas
Android development. Analysis and development of embedded software for remote testing units of passive optical networks in Java.

2011 **Adjunct Professor**, *EEP – Piracicaba Engineering School*, Piracicaba
Taught Analytic Geometry and Linear Algebra.

Education

2008–2010 **MSc. in Philosophy**, *Unicamp – Campinas State University*, Campinas
In the area of Mathematical Logic, with a dissertation on logic programming.

2002–2007 **Computer Engineer**, *Unicamp*, Campinas

Technical highlights

Clean Code	DDD, CQRS, TDD
Agile	Scrum, Kanban
Observability	APM, (E2E) Distributed Tracing
Parallel/concur./distr.-systems development	LMAX Disruptor, io_uring, libuv
OO Design Patterns	RESTful APIs development
C/C++, .NET(C#), Java, Node.js	Boost, Asio, RAII/C++ Patterns
GDB, Valgrind, strace	JUnit 5, NUnit, xUnit, Catch2
Datadog, Grafana, Prometheus	JMeter, SonarQube, Wireshark
FIX/FAST/SBE, RabbitMQ, Kafka	Postgres, SQL Server, MongoDB
Bash/BAT/PowerShell scripting	CMake, Conan, vcpkg, NuGet, Gradle
Docker, K8s, NixOS/Nix	GitHub Actions, ArgoCD, Jenkins
GCP, AWS	OAuth, JWT
B3's UMDF Platform	Payments APIs (PIX)

Languages

English Advanced/Fluent.
Spanish Conversational.
Portuguese Native.